

expand

August 22, 2026

This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
Copyright (C) 2010–2023 Joel Andersson, Joris Gillis, Moritz Diehl,
KU Leuven. All rights reserved.
Copyright (C) 2011–2014 Greg Horn

CasADi is free software; you can redistribute it and/or
modify it under the terms of the GNU Lesser General Public
License as published by the Free Software Foundation; either
version 3 of the License, or (at your option) any later version.

CasADi is distributed in the hope that it will be useful,
but WITHOUT ANY WARRANTY; without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU
Lesser General Public License for more details.

You should have received a copy of the GNU Lesser General Public
License along with CasADi; if not, write to the Free Software
Foundation, Inc., 51 Franklin Street, Fifth Floor, Boston, MA 02110-1301 USA

1 expand

```
[1]: import casadi as ca  
import casadi as c
```

We construct a simple MX expression

```
[2]: x = ca.MX.sym("x",2,2)  
y = ca.MX.sym("y",2,1)
```

```
[3]: z = x @ y
```

Let's construct an MXfunction

```
[4]: f = ca.Function("f", [x,y],[z])
```

We expand the MX expression into an SX expression

```
[5]: fSX = f.expand('fSX')
```

```
[6]: print("Expanded expression = ", fSX.str(True))
```

```
Expanded expression = fSX:(i0[2x2],i1[2])->(o0[2]) SXFunction
Algorithm:
@0 = input[0][0];
@1 = input[1][0];
@0 = (@0*@1);
@2 = input[0][2];
@3 = input[1][1];
@2 = (@2*@3);
@0 = (@0+@2);
output[0][0] = @0;
@0 = input[0][1];
@0 = (@0*@1);
@1 = input[0][3];
@1 = (@1*@3);
@0 = (@0+@1);
output[0][1] = @0;
```

2 Limitations

Not all MX graphs can be expanded. Here is an example of a situation where it will not work.

```
[7]: linear_solver = ca.Linsol("linear_solver", "csparse", x.sparsity())
g = linear_solver.solve(x, y)
G = ca.Function("G", [x,y], [g])
```

This function cannot be expanded.

```
[8]: try:
      G.expand('G_sx')
except Exception as e:
    print(e)
```

```
Error in Function::call for 'G' [MXFunction] at
.../casadi/core/function.cpp:523:
.../casadi/core/linsol_internal.cpp:65: eval_sx not defined for CsparseInterface
```